Rabiya Tahir

Information Technology University, Lahore

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Education

Information Technology University Sep 2023 – May 2027
BS Artificial Intelligence | PEEF Scholarship Recipient *Lahore*

Kinnaird College 2021 – 2023
Intermediate (ICS) – 89% *Lahore*

Experience

Turing (via Swift Solutions) Oct 2024 – Present
LLM Trainer (QA) & LLM Systems Developer *Remote*

- **AI Trainer & Prompt Engineer (Anthropic)** Evaluated LLM code generation for logical consistency, security compliance, and task accuracy; executed and analyzed prompt-response cycles using GitHub-based tooling; contributed to RLHF pipelines for improving model performance.
- **NVIDIA Multi-App Task QA** Audited single-app and multi-app LLM task prompts against a 33-rule compliance framework, evaluating execution logs for accuracy; processed ∼16 tasks/day.
- **LLM Systems Developer (DB Domain)** Designed domain-specific interaction frameworks for LLM agents: structured **policies** (e.g., GitHub workflows), **tools/APIs** (`fetch_repo`, `create_repo`), and a collaborative **database schema**.
- **Data Annotator (Meta)** Annotated structured video datasets and generated metadata used in downstream LLM training pipelines.
- **Agent Evaluator (Meta – CUA)** Evaluated computer-use agents by testing real-world task execution, edge cases, and output correctness.

Information Technology University – Database Management System Feb 2025 – June 2025
Teaching Assistant *Lahore*

- Mentored 40+ undergraduates in relational database design, SQL, and query optimization; held office hours and graded assignments/exams.

CodeRush 2026 ML Competition – 3rd Place | GDG ITU 2026

- Built economic class classifier (Lower/Middle/Upper) on bag-structured demographic data using LightGBM + Random Forest ensemble with threshold tuning.
- Achieved Macro F1: 0.7325; engineered 198 bag-level features from 29 raw columns; improved middle-class recall by +46% vs logistic regression baseline.

Projects & Competitions

Visual Prompt Injection Defense for Vision-Language Models | Python, PyTorch, FastAPI, Docker Ongoing FYP, 2026 – 2027

- Designing a dual-layer guardrail that screens images for embedded/injected instructions before they reach a vision-language agent, preventing attackers from hijacking agent actions via manipulated images.
- Benchmarked three Layer 1 image filters (FFT, PatchCore CVPR 2022, WinCLIP CVPR 2023) on CPU across 4 datasets, using paired clean/injected controls to separate true text detection from image-type mismatch.
- PatchCore (photo memory bank) detected embedded text at AUROC 0.970 (blank-band control 0.501) vs. 0.28% for FFT at 10% FPR; zero-shot WinCLIP reached F1 0.76 on 940 images, beating a 3-method ensemble (F1 0.72).

Live Wallpaper Dashboard | Electron, JavaScript, Win32 API, Google Calendar API 2026

- Built and shipped a Windows app rendering a live dashboard (clock, weather, calendar, alarms) behind desktop icons, with natural-language alarm parsing and Google Calendar OAuth sync; auto-updating releases via GitHub Actions CI/CD.

Local Autonomous Tool-Use Agent | Python, Ollama, Llama 3.2 2026

- Deployed fully local, air-gapped LLM agent via Ollama (3B model), eliminating external API dependency for agentic workflows.
- Built multi-turn agent loop with custom JSON function-calling to autonomously generate 4-step automation plans, removing manual data-entry bottleneck.

Prompt Injection Guardrail | Python, FastAPI, React, Docker 2026

- Full-stack guardrail layer using dual-layer FastAPI pipeline (heuristics + LLM judge) to detect prompt injection attempts before reaching agent context.
- Eval Dashboard computing Precision, Recall, F1 against synthetic attack vectors; containerized with Docker and GitHub Actions CI/CD.

XAI Medical Decision Support System | Python, scikit-learn, SHAP, FastAPI, Streamlit 2026

- Explainable AI system on Heart Disease and Adult Income datasets; trained 4 models with SHAP for global/local prediction explanations.
- Led fairness analysis auditing demographic parity, equal opportunity, and disparate impact across sex and age groups; served via FastAPI + Streamlit.

Leadership

GDG ITU | *Management Head* 2024–2025

- Organized and managed CodeRush 2025, a campus-wide ML competition.

EMS ITU | *Vice President* 2024–2025

- Managed society operations and led events including bonfires and welcome parties.

Technical Skills

Languages: Python, C/C++, SQL, JavaScript, HTML/CSS
Tools: Ollama, REST APIs, Git, MySQL
AI/ML: LLMs, RLHF, Model Evaluation, Agentic Systems, PyTorch, HuggingFace